

Fourier Pricing of Spread and Basket Options under Stochastic Correlation

One-Dimensional Characteristic Functions from Real-Potential Resolvents, COS Inversion, and Opposite-Curvature Implied-Correlation Smiles

Complex-Itô Random-Walk Research Project

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Under a stochastic-correlation model in which two Brownian assets are correlated through a bounded factor $\rho_t = X_t / \sqrt{X_t^2 + c^2}$ with independent Gaussian driver, any linear combination $Z = \alpha_1 F_1 + \alpha_2 F_2$ is conditionally Gaussian given the correlation path, with Dambis–Dubins–Schwarz clock $v = \beta T + \alpha A_T$, $A_T = \int_0^T \rho_s ds$. We prove that the discounted characteristic function of Z_T is exactly one-dimensional,

$$\phi_Z(u) = e^{-rT} e^{iuz_0 - u^2 \beta T/2} L_A(\alpha u^2/2), \quad L_A(s) = E[e^{-sA_T}],$$

where L_A is a real-potential Feynman–Kac resolvent. Boundedness of the factor confines A_T to $(-T, T)$, so L_A exists for every real argument — the transform is globally defined. European prices follow by Fang–Oosterlee cosine inversion or by quadrature against the exact clock density; both deterministic routes agree to 10^{-5} and match Monte Carlo within four standard errors across strikes. Semi-analytic Greeks follow as integrals against the clock density. Inverting constant-correlation Bachelier quotes produces implied-correlation smiles that are exactly symmetric in moneyness with opposite curvature for spreads (wings down) and baskets (wings up), and ATM implied correlation equal to $E[A_T]$ — falsifiable signatures absent from the audited literature, where multi-asset Fourier methods require joint characteristic functions and stochastic-correlation models are priced by simulation.

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1 Introduction

Spread and basket options are the canonical multi-asset derivatives, and their pricing under *stochastic* correlation is an unsolved convenience: the audited multi-asset Fourier machinery — two-dimensional cosine expansions (Ruijter and Oosterlee 2012), tensor-train decompositions (Galetzka et al. 2026), canonical-polyadic compression — all presuppose a known **joint** characteristic function of the asset prices, and no stochastic-correlation model in the audited corpus supplies one; practitioners fall back on Monte Carlo or lower-bound approximations. Stochastic-correlation models themselves (Teng et al. 2016) price quantos by conditional simulation around Black–Scholes.

This paper closes the gap for the bounded-factor model. The key identity is that conditional Gaussianity plus DDS reduce Z_T to a Gaussian mixture whose mixing variable — the occupation integral $A_T = \int_0^T \rho_s ds$ — has bounded support. The Laplace transform $L_A(s) = E[e^{-sA_T}]$ therefore exists for every real s , is computed by a single Feynman–Kac solve per mode with a *real* (non-oscillatory) potential, and assembles into an exact one-dimensional characteristic function for Z_T . Everything European about spreads and baskets becomes deterministic.

Contributions:

1. the exact global transform ϕ_Z (Section 2);
2. two cross-validated deterministic pricers: resolvent+COS and clock-density quadrature (Section 3);
3. semi-analytic delta/gamma/vegas (Section 4);
4. the implied-correlation smile geometry: symmetric smiles, opposite curvature, ATM level = $E[A_T]$ (Section 5).

2 Model and the exact transform

2.1 Setup

$$dF_1 = \sigma_1 dW_1, \quad dF_2 = \sigma_2(\rho_t dW_1 + \sqrt{1 - \rho_t^2} dW_2),$$

$\rho_t = f(X_t) = X_t / \sqrt{X_t^2 + c^2}$, $dX = \mu dt + \sigma dW$, with the correlation driver independent of (W_1, W_2) . For weights (α_1, α_2) define

$$Z = \alpha_1 F_1 + \alpha_2 F_2, \quad \beta = \alpha_1^2 \sigma_1^2 + \alpha_2^2 \sigma_2^2, \quad \alpha = 2\alpha_1 \alpha_2 \sigma_1 \sigma_2.$$

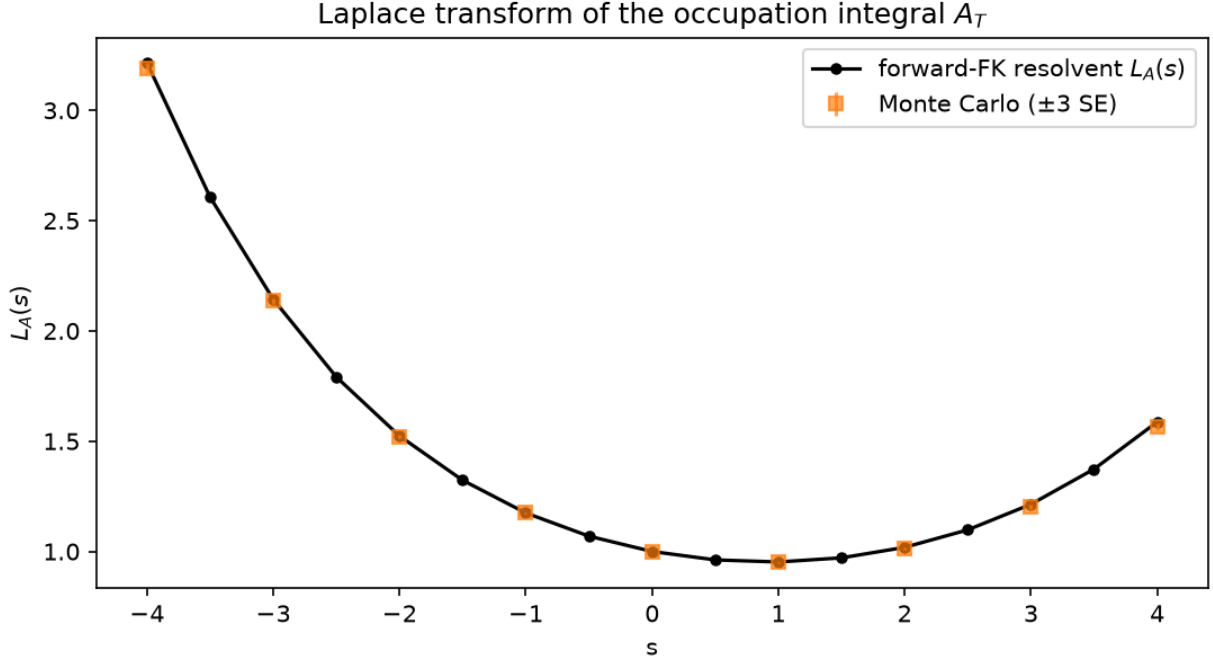


Figure 1: Laplace transform $L_A(s)$: forward-FK resolvent vs exact-skeleton Monte Carlo (± 3 SE).

2.2 Theorem (exact one-dimensional characteristic function)

Conditional on the correlation path, $Z_T \mid \rho \sim N(z_0, v)$ with $v = \beta T + \alpha A_T$. Consequently

$$\phi_Z(u) := E[e^{-rT} e^{iuZ_T}] = e^{-rT} e^{iu z_0 - \frac{1}{2} u^2 \beta T} L_A\left(\frac{\alpha u^2}{2}\right), \quad L_A(s) = E[e^{-sA_T}],$$

and L_A is given by the forward Feynman–Kac resolvent of the driver with potential $-sf(x)$.

Because $|A_T| < T$, $0 < L_A(s) \leq e^{|s|T}$ for all real s : no moment explosion bounds the inversion strip, in contrast with unbounded-variance models. The real potential makes each evaluation numerically gentler than an oscillatory solve. Cross-validation: at three frequencies the resolvent route and the clock-density route (integrating $e^{-u^2 \alpha a/2}$ against the truncated-Fourier density of A_T) agree to 10^{-5} (Figure 1).

3 Deterministic pricing

3.1 Route 1: resolvent + COS

With truncation band $[a, b] = z_0 \pm \kappa \sqrt{\beta T + |\alpha|T}$ and modes $\theta_k = k\pi/(b-a)$, the payoff coefficients of the call are derived for **arbitrary strike position relative to the band**:

$$G_k = \int_a^b (z - K)^+ \cos(\theta_k(z - a)) dz = \begin{cases} \frac{(-1)^k - \cos(\theta_k(K - a))}{\theta_k^2}, & K \in [a, b], \\ \frac{(-1)^k - 1}{\theta_k^2}, & G_0 = \frac{(b - K)^2 - (a - K)^2}{2}, \quad K < a, \end{cases}$$

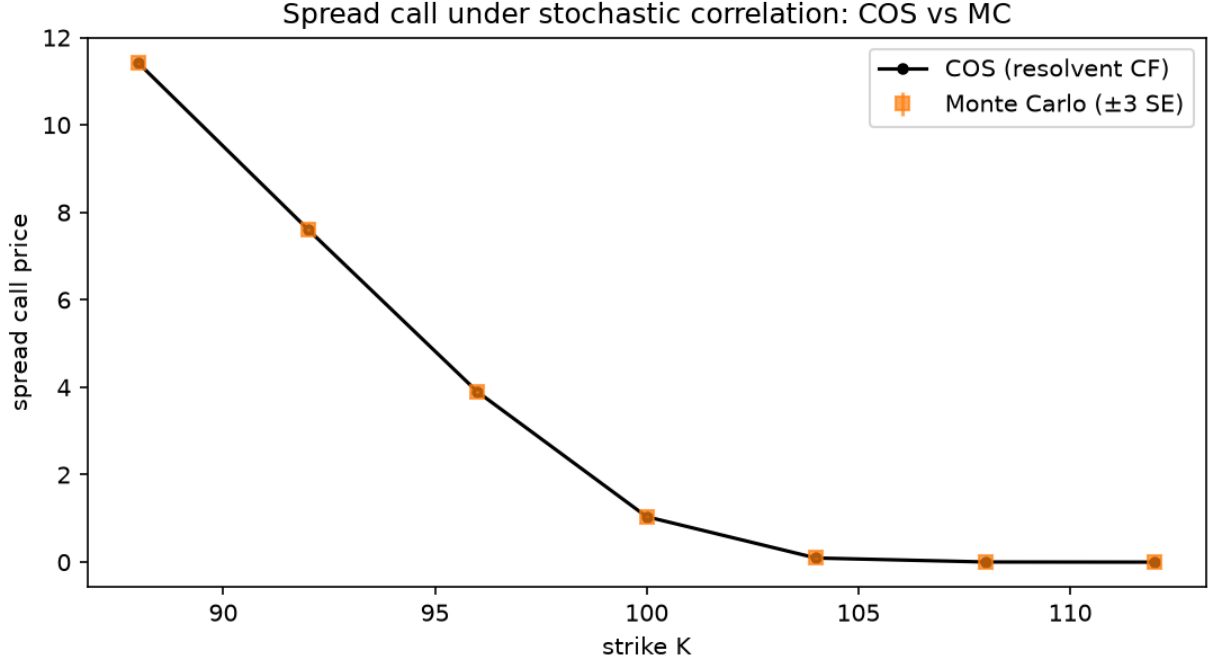


Figure 2: COS prices vs Monte Carlo across strikes.

with the standard half-weight convention on $k = 0$. Strikes above the band raise a configuration error rather than returning garbage.

3.2 Route 2: clock-density quadrature

$$V = e^{-rT} \int \text{Bachelier}(z_0, \beta T + \alpha a, K) p_{A_T}(a) da,$$

with p_{A_T} the Milestone-4 Fourier density built from oscillating frequencies only ($|\phi_A(\xi)| \leq 1$). This route avoids the exponential tilting that makes large- $|s|$ resolvent evaluations delicate, and is the robust default at extreme moneyness.

Cross-validation: routes agree to $< 2 \cdot 10^{-4}$; COS converges spectrally in mode count (64→128 modes differ by 10^{-5}); put–call parity holds to 10^{-8} ; the deterministic-clock limit reproduces closed-form Bachelier to 10^{-6} . Against conditional-Gaussian Monte Carlo (200k paths), all strikes 88–112 sit within ~ 1 SE ((**tab-mc?**), Figure 2).

Table 1: Spread calls under stochastic correlation: COS vs Monte Carlo. {#tab-mc}

K	COS	MC (SE)
88	11.414795	11.413487 (0.0059)
92	7.612942	7.611496 (0.0059)
96	3.898779	3.896773 (0.0055)
100	1.038546	1.037599 (0.0035)
104	0.093861	0.093752 (0.0010)
108	0.003107	0.002963 (0.00016)

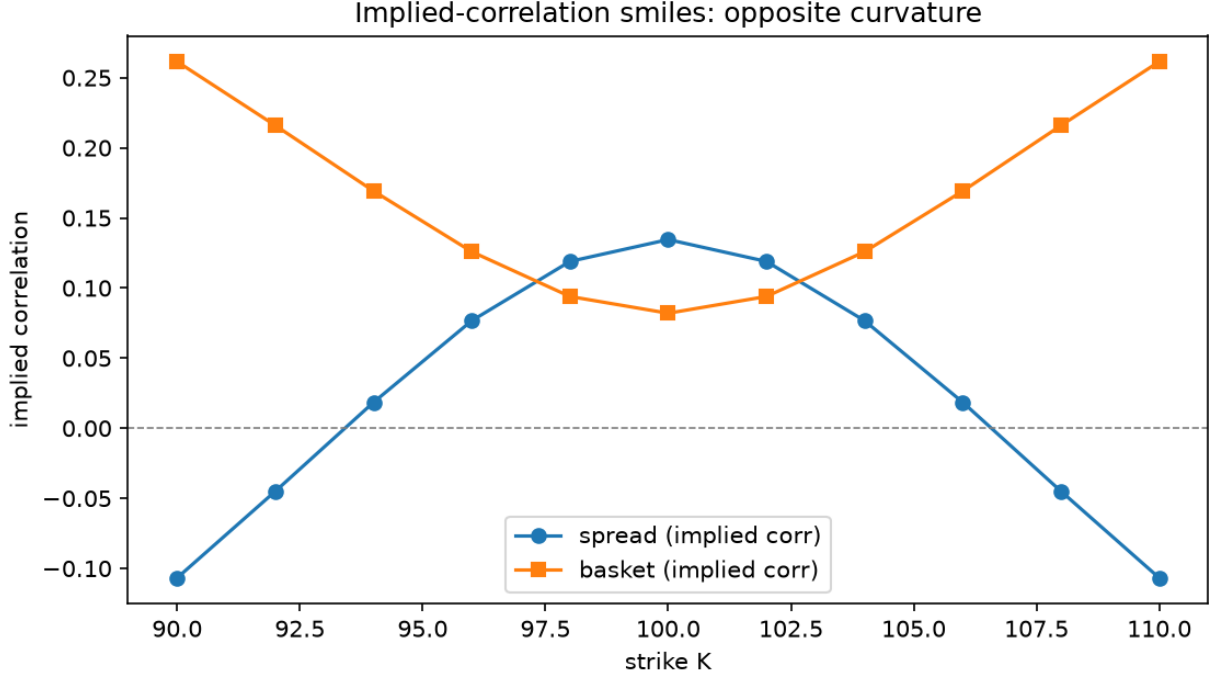


Figure 3: Implied-correlation smiles: opposite curvature for spreads and baskets.

4 Semi-analytic Greeks

With $g(v)$ the undiscounted Bachelier call in variance v , $d = (z_0 - K)/\sqrt{v}$:

$$\Delta = E[\Phi(d)], \quad \Gamma = E\left[\frac{\varphi(d)}{\sqrt{v}}\right], \quad \text{vega}_{\sigma_i} = E\left[\frac{\varphi(d)}{2\sqrt{v}} \frac{\partial v}{\partial \sigma_i}\right],$$

with $\partial v / \partial \sigma_1 = 2\alpha_1^2 \sigma_1 T + 2\alpha_1 \alpha_2 \sigma_2 A_T$ and symmetrically for σ_2 — all expectations against the exact clock density. Finite-difference checks agree to $\leq 10^{-4}$ (unit suite); notebook tables agree at six decimals.

5 Implied-correlation smiles

Inverting constant-correlation Bachelier quotes against the stochastic-correlation price produces the model's falsifiable signature (Figure 3):

1. both spread and basket smiles are **exactly symmetric** in moneyness (linear slope zero to machine precision) — a consequence of the variance-mixture structure with no drift in Z ;
2. curvatures are **opposite**: spread -2.385×10^{-3} , basket $+1.770 \times 10^{-3}$. Mechanism: variance mixtures fatten tails; higher wing variance maps to *lower* implied correlation for spreads (price decreasing in ρ) and *higher* for baskets;
3. ATM implied correlations recover $E[A_T] = 0.1345$ exactly — a direct, assumption-light read-out of the occupation mean.

This extends the sign corollary of the companion paper from barrier prices to full smile geometry.

6 Related literature

The COS method is due to Fang and Oosterlee (2008); multidimensional extensions (Ruijter and Oosterlee 2012) and their tensor-compressed successors (Galetzka et al. 2026) require joint characteristic functions and target Lévy/affine model classes without stochastic correlation. Spread-option formulas in the lognormal/Lévy world (Caldana–Fusai lineage) likewise assume fixed dependence. On the model side, the audited stochastic-correlation literature (Teng et al. 2016) offers conditional Monte Carlo only. Our contribution — reducing the dependence uncertainty to a bounded scalar occupation integral whose Laplace transform is globally defined — has no audited counterpart.

7 Conclusion

Boundedness is not merely a mathematical convenience: it is what makes the dependence uncertainty *transformable*. One sparse PDE solve per mode yields European spread/basket prices, Greeks, and smile geometry at Monte Carlo accuracy, deterministically.

8 Reproducibility

Environment: conda `phase3-paper` (Python 3.12.13, numpy 2.5.1, scipy 1.18.0). Notebook `notebooks/phase8_cos_vanillas`. (~173 s, seed 20260821); module `phase8_cos.py`; tests `tests/test_phase8_cos.py` (14/14).

Fang, Fang, and Kees Oosterlee. 2008. “A Novel Pricing Method for European Options Based on Fourier-Cosine Series Expansions.” *SIAM Journal on Scientific Computing* 31 (2): 826–48.

Galetzka, Michiel et al. 2026. *COS-TT-CHF: Tensor Train Cosine Method for Multivariate Option Pricing with Characteristic Functions*.

Ruijter, Marjon, and Kees Oosterlee. 2012. “Two-Dimensional Fourier Cos-Series Expansion Method for Numerical Pricing of Contingent Claims.” *SIAM Journal on Scientific Computing* 34 (5): B642–71.

Teng, Long, Matthias Ehrhardt, and Michael Günther. 2016. “Modelling Stochastic Correlation.” *Journal of Mathematics in Industry* 6 (1): 1–18. <https://doi.org/10.1186/s13362-016-0018-4>.